A-A-50852A
13 December 1985

COMMERCIAL ITEM DESCRIPTION

MICROSCOPE, OPTICAL, BINOCULAR

This commercial item description shall be used in preference to Military Specification MIL-M-37869.

This commercial item description covers a revolving nosepiece, inclined binocular microscope with planachromatic objectives and built-in light source, suitable for histopathological and pathological studies.

Salient characteristics:

As a minimum, the microscope shall consist of a base with variable light source, stand with supporting arm and focusing controls, inclined binocular body (head), revolving nosepiece assembly with three planachromatic objectives and dust plug(s), stage mount, mechanical stage, and substage condenser. The focusing of the microscope shall be controlled by the vertical movement of the microscope stage or the objective.

The base shall have built-in, a variable light source (illuminator) for operation from 115 volt, 50/60 Hz, AC supply lines through a power cord. The light source shall be a 6 volt, 20 watt tungsten halogen lamp. The circuitry for the light source shall include a continuous voltage (intensity) control, on-off switch and transformer. The lamp shall be supported in a detachable lamp socket for easy replacement. The power cord shall be a flexible 3-conductor cord that terminates with a non-detachable 3-blade male plug. The illuminator shall have a filter holder with a removable blue filter and a built-in field diaphragm for Koehler illumination.

The stand shall house the focusing mechanism, and shall be attached to the base. The supporting arm of the stand shall support the binocular body. The focusing mechanism shall raise and lower the nosepiece assembly or stage. The focusing mechanism shall include coarse and fine controls that are manually controlled by coarse and fine knobs (dials) located in the lower position on both sides of the stand. For purposes of measuring specimen thickness, the fine adjustment shall be graduated in at least two (2) micrometer divisions with one complete revolution representing 100 micrometers as a minimum. For protecting an objective and specimen (slide), suitable means shall be included in the stand to limit the travel of the coarse and fine focusing controls.

The binocular body (head) shall have eyepiece tubes with 10X wide field eyepieces and eyeguard relief. The eyepiece tubes shall be inclined 30-40 degrees from the horizontal. The interpupillary distance (between eyepiece tubes) shall be adjustable within a range that includes 56 to 72 millimeters (mm). The interpupillary adjustment shall include a millimeter scale for easy reference. Automatic tube length compensation shall be included if the tube length is affected by changes in the interpupillary settings. The eyepieces or eyepiece tubes shall include provisions for installing micrometer discs. The eyepieces or eyepiece tube(s) shall be focusable for acuity equalization of a user's eye. The eyepieces shall have at least an 18 mm diameter field of view and a minimum eye relief distance of 17 mm. The binocular body shall be 360 degrees rotatable on the supporting arm (stand). The binocular body shall also be detachable for interchangeability with heads (e.g. a trinocular body for photomicrographic procedures).

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Three objectives shall be supplied with a quadruple or quintuple 360 degree revolving nosepiece having a ball detent index. Dust plugs shall be included in the unused objective hole(s) of the nosepiece. The objectives furnished shall be 10X, 40X or 45X, and 100X oil immersion planachromatics having numerical apertures of 0.25, 0.65 or 0.66, and 1.25 respectively. The objectives shall be color coded and marked for identification. In the nosepiece, the objectives shall be parfocal and parcentered so they are not affected by switching from one objective to another. The 40X, 45X and 100X objectives shall be retractable (i.e. spring loaded) when contact is made with a microscope slide or its cover slip. As an alternate to retractable objectives, the nosepiece assembly shall be retractable when an objective makes contact with a microscope slide or its cover slip.

The stage mount shall accommodate microscope slides that include 25 by 75 mm and 50 by 75 mm sizes. The stage mount shall include a slide holder that is detachable by releasing a thumbscrew hold. The slide holder shall be in the form of a finger assembly that can grasp the sides of a microscope slide instead of its corners. The finger assembly shall also be retractable to allow more space on the stage mount for manual scanning of a microscope slide. The aperature width of the stage mount shall be not less than 20mm. A right hand ungraduated mechanical stage, as a minimum, shall control the horizontal movement of the stage mount. The mechanical stage shall have ball bearing slideways in two perpendicular directions that can give complete coverage under an objective of both size microscope slides.

The substage condenser shall be located below the mechanical stage in a fork-type or dovetail mount which is moved vertically by rack and pinion control for focusing purposes. The substage condenser shall have a numerical aperature (N.A.) of 1.25 or 1.30, and shall include an integral iris diaphragm and shall provide an even full field illumination and a variable aperature control for all the objectives, i.e. from 10X to 100X (oil immersion). The fork shall contain two knurled centering screws. The condenser shall also be removable from the mount without having to remove the centering screws.

Provisions shall be made in the microscope to control the vertical movement of the condenser so that it does not make contact with the stage mounted microscope slide or the base mounted light source. In addition, a quick set condenser shall be supplied which allows the condenser to be lowered from a focused position and then returned to its original position by being raised to a positive stop.

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The microscope shall be capable of accepting the following accessories which shall be commercially available from the manufacturer, but are not required to be supplied with the microscope:

1. A rotatable polarizer to fit the light source and an associated slide attachment to fit below the binocular body (head).

2. A paraboloid or oil immersion dark field substage condenser that is interchangeable with the substage condenser that is supplied with the microscope.

3. A dual viewing adapter with a built-in pointer, for use with the binocular body (head) that is supplied with the microscope.

4. A trinocular head for photomicroscopy, that is interchangeable with the binocular body (head) that is supplied with the microscope.

To eliminate vibrations during photomicroscopy, the fully assembled microscope (i.e., stand, accessory trinocular body with objectives, stage mount and substage condenser) shall not weigh less than 19 pounds.

The microscope shall be legibly and permanently marked with the name or registered trademark of the manufacturer, serial number and model number.

The microscope shall be supplied with a protective dust cover.

Workmanship. The microscope shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). Each microscope, as specified, constitutes one unit.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

The following National Stock Number is covered by this document:

6650-01-158-8014

Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITY:
VA-OSS

51618

Project No. 6650-0131

Location: S3-10-3

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